

Year 10 Practice test

① scores on a recent test:

54, 63, 91, 67, 63, 82, 81, 60, 79, 67, 79,
65, 83, 69

Sort the data $\rightarrow$ 54, 60, 63, 63, 65, 67, 67, 69, 79, 79, 81, 82, 83, 91

$$\text{min} = 54$$

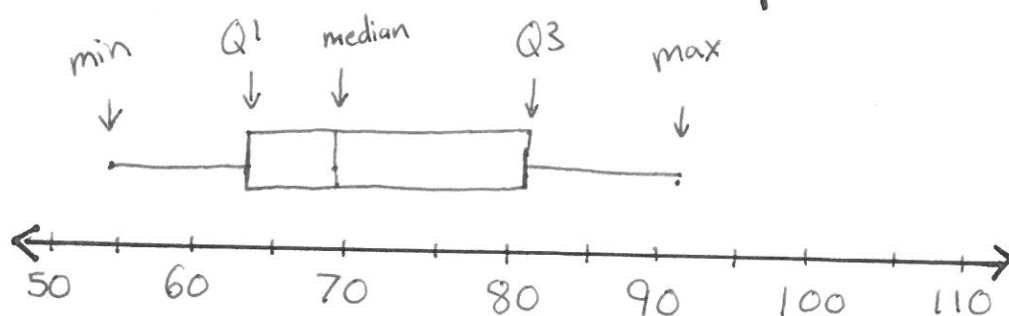
$$Q1 = 63$$

$$Q2 (\text{median}) = 68$$

$$Q3 = 81$$

$$\text{max} = 91$$

Draw box + whisker plot



② Consider the given scores. Within what range do the middle 50% of scores lie?

10, 11, 12, 13, 14, 15, 15, 16, 17, 18, 19, 20

between 12.5 and 17.5

* this is also called the IQR - Inter-Quartile Range.

③ Define outlier: a number significantly far away from the median

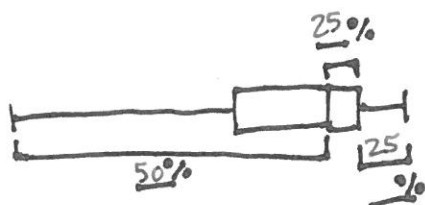
How do you calculate?

$$Q3 + (1.5 \times \text{IQR})$$

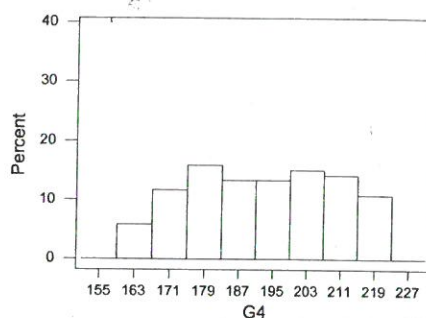
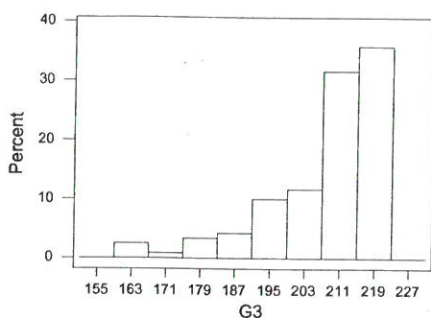
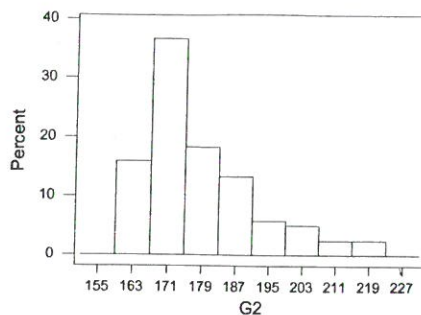
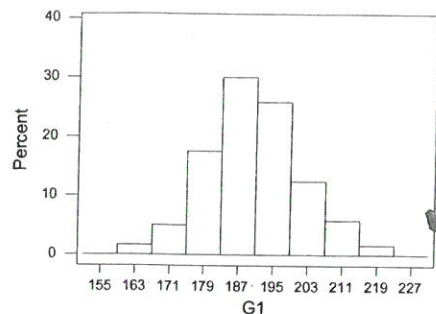
or

$$Q1 - (1.5 \times \text{IQR})$$

④ Label percentages on the box + whisker plot:



Match the histograms to boxplots

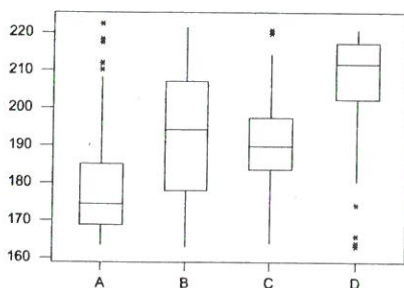


G1: C

G2: A

G3: D

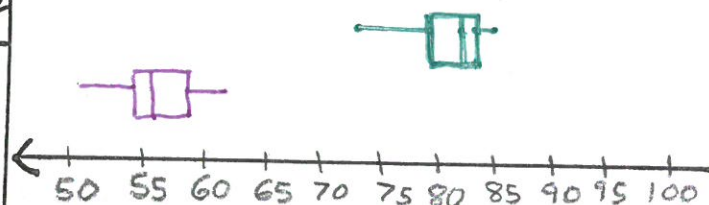
G4: B



Median age at death in various states of Australia
construct a parallel box+whisker plot

Indigenous females	Non-Indigenous females
58, 62, 54, 55, 51, 56, 57	83, 84, 79, 81, 79, 83, 82
min 51	min 74
Q1 54	Q1 79
med 56	med 82
Q3 58	Q3 83
max 62	max 84

≡ non-indigenous
≡ indigenous



Comment on the differences in the data:

There are significant differences in the data for median age at death comparing non-indigenous and indigenous females. The indigenous data has a min of 51 y.o. and a max of 62 y.o. which is below the min of 74 y.o. for non-indigenous females. They have similar ranges (around 9) and the same I.Q.R. of 4.